

Downside dominance in industry volatility networks: The role of shock classification

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Abstract

Using 30 U.S. industries, we show that the apparent disappearance of downside dominance after factor adjustment partly reflects a change in what counts as a negative shock. In 2014–2025, negative-return days account for 94.37% of positive predictive association strength in the raw-return network. Market adjustment lowers this share to 63.64% when the original signs are retained and to 48.95% when residual signs define the channels. Relabeling therefore accounts for 32.4% of the measured reduction; three- and five-factor models give similar results. The ordering survives alternative transformations and treatment of zero observations. Simulations without direct links between industry-specific shocks can generate strong raw downside dominance, but observed residual strength and fixed-sign imbalance exceed this benchmark. Factor adjustment thus attenuates the imbalance; near balance also depends on shock classification and does not imply that residual dependence has disappeared.

Keywords: Downside risk; industry networks; common factors; volatility; shock classification.

1 Introduction

A negative return and a negative factor-adjusted return describe different events. An industry can lose value while outperforming the return implied by a falling market. Its raw return is negative, but its residual is positive. A comparison of raw and residual volatility networks therefore changes both the return magnitudes and the days classified as adverse. Overlooking this distinction can make a change in measurement appear to be the disappearance of downside dependence.

Signed variation is informative about future volatility (Barndorff-Nielsen et al., 2008; Patton and Sheppard, 2015), and negative and positive volatility can have different connectedness patterns (Baruník et al., 2016). Common factors also account for an important component of measured financial connectedness (Miao et al., 2023). We connect these insights by asking how much apparent downside dominance survives factor adjustment when the definition of a negative day is held fixed. We measure weekly variation from daily returns, whereas much of this literature uses prices observed within the trading day.

The contribution is a simple intermediate comparison. We first remove factor exposure while retaining each industry’s original daily signs. We then change only the signs used to divide residual

variation. The first step sharply reduces downside dominance; the second brings the measured share close to one-half. For market adjustment, changing labels accounts for almost one-third of the total reduction.

Two checks sharpen the interpretation. The ordering survives several ways of transforming the data and handling weeks with no observations in one sign category. Simulations also show that a nearly balanced residual network can retain more strength than a model without direct links between industry-specific shocks predicts. Thus attenuation, balance, and disappearance of residual dependence are distinct claims.

2 Data and measurement

We use the 30 value-weighted U.S. industry portfolios and daily factors from French (2026). We use the July 2026 data release, retrieved in September 2026, with observations after December 31, 2025 excluded. Weekly data extend from 1972 to 2025; the reported comparison covers 2014–2025, using earlier observations for estimation. This is a fixed retrospective release; successive historical releases are not reconstructed.

Market, Fama–French three-factor (FF3), and five-factor (FF5) models remove exposure to common returns (Fama and French, 1993, 2015). Each industry regression uses the preceding 504 trading days and is updated every 21 days. Its coefficients are estimated before application, but the adjustment uses realized contemporaneous factors. It is therefore an after-the-fact return decomposition.

Weekly variation sums squared daily returns, expressed as decimals, in Friday-ending weeks. For adjusted returns, the squared residuals are divided in two ways. *Original signs* assign a day to the negative category whenever its raw industry return is negative. *Residual signs* assign it there whenever its factor-adjusted return is negative. The remaining variation forms the nonnegative category. Zeros belong to that category. The measure omits intraday variation.

The network relates variation in one industry to the next week’s total variation in another. It controls for the receiving industry’s recent volatility and lagged market volatility. The controls summarize recent volatility over one, four, and thirteen weeks, adapting Corsi (2009). Each industry pair and sign category is assessed separately. The relationships are estimated over 156 response weeks and updated every four weeks, giving 156 distinct estimates in 2014–2025. These inherited choices keep comparisons aligned; their optimality is not claimed. Appendix A gives the equations.

For each sign category, network strength is the average positive correlation remaining after the controls; negative correlations are set to zero and self-links are excluded. The *negative share* is negative-category strength divided by combined strength across both categories. We average this share across update dates. A value above 50% indicates stronger positive associations from negative-day variation. It is neither a share of return variance nor a causal transmission estimate.

Holding original signs fixed changes return magnitudes, receiving-industry outcomes, and their volatility controls together. The subsequent move to residual signs changes only category assignments. The resulting breakdown describes this sequence of measurement choices, without separating economic shocks into independent causal components.

3 Results

3.1 Factor adjustment reduces dominance; changing signs brings balance

The raw-return network has a mean negative share of 94.37%. Subtracting only the risk-free rate leaves it almost unchanged, at 94.20% using excess-return signs. Market adjustment lowers the share to 63.64% when original signs remain fixed, and to 48.95% when residual signs determine the categories. FF3 and FF5 produce similar results (Table 1). Market adjustment therefore captures most of the change, but near balance additionally depends on how adverse days are defined.

Table 1: Negative shares after factor adjustment, 2014–2025

Factor model	Original daily signs	Residual daily signs
Market	63.64 [62.40, 64.91]	48.95 [47.26, 50.67]
FF3	62.76 [61.21, 64.31]	49.16 [48.19, 50.18]
FF5	62.84 [61.11, 64.57]	49.14 [47.94, 50.32]

All entries are percentages; brackets show conditional 95% intervals. The raw-return share is 94.37% [92.57, 96.17]. Shares average 156 date-specific ratios, not ratios of mean strengths. Intervals resample already fitted networks; see Appendix A.

For the market model, adjustment with original signs reduces the share by 30.72 percentage points, with a conditional 95% interval of [29.81, 31.66]. Changing signs reduces it by another 14.70 points, with interval [13.27, 16.10]. Relabeling accounts for 32.4% of the 45.42-point total decline. These changes use unrounded estimates and need not equal differences between displayed rounded shares.

This is more than a change in the aggregate amount of negative variation. Market adjustment changes the sign on 26.19% of industry-days. Switching from original to residual signs increases the pooled negative share of squared residuals slightly, from 48.44% to 48.88%, while substantially reducing the network’s negative share. Reclassification reorganizes variation across industries and weeks even when the aggregate split barely changes.

The result also depends on counting only positive associations. Before negative correlations are discarded, raw correlations average 0.1084 for the negative category and -0.0971 for the nonnegative category. The latter’s small positive strength therefore does not imply an absence of association. Downside dominance here concerns the positive part of the measured relationships.

3.2 The ordering survives alternative data treatments

We change how zero values enter logarithms, replace magnitudes with ranks, and exclude weeks lacking either sign category while matching retained observations across specifications. Across eight treatments, raw negative shares range from 93.01% to 96.34%. Factor-adjusted shares remain between 58.25% and 63.64% with original signs, and between 48.57% and 51.35% with residual signs. Appendix B reports the full comparison.

The pattern therefore survives the tested choices for scaling the data and handling zeros. Its exact magnitude varies, especially when original signs are retained. Excluding zero weeks also changes the estimation sample. These checks support the ordering of results, rather than identical estimates under every transformation.

3.3 Near balance does not imply that only noise remains

Positive estimated correlations can arise by chance. We therefore compare observed networks with simulated economies that retain fitted common returns, allow each industry’s volatility to persist and respond differently to gains and losses, but draw industry-specific shocks independently. Factor regressions and networks are re-estimated in each of 499 simulations per factor model. Appendix C specifies the benchmark.

Simulated raw negative shares average 94.70–96.39% across generating models. Strong raw downside dominance can therefore arise without direct links between industry-specific shocks. The observed share falls within the FF3 and FF5 simulation ranges, while the market model generates greater dominance. This benchmark does not reproduce every raw-network statistic, including total strength under FF3 and FF5.

Observed residual strength nevertheless exceeds the benchmark (Table 2). Across all 24 reported residual-strength comparisons —three factor models, two sign rules, and four transformations—the observed statistic exceeds every corresponding simulation. All 12 original-sign residual-share comparisons do so as well. The associated one-sided simulation probability is 0.002, the smallest value resolved by this experiment; these probabilities are unadjusted for multiple comparisons.

Table 2: Residual networks compared with the independent-shock benchmark

Factor model	Observed	Simulation mean	95% simulation range
<i>A. Combined strength using residual signs</i>			
Market	0.07043	0.06351	[0.06083, 0.06631]
FF3	0.07859	0.06319	[0.06043, 0.06582]
FF5	0.07536	0.06349	[0.06113, 0.06638]
<i>B. Negative share using original signs (%)</i>			
Market	63.64	55.02	[51.10, 59.13]
FF3	62.76	53.88	[49.85, 58.32]
FF5	62.84	54.40	[49.88, 58.25]

Baseline transformation; 499 simulations for each generating model. Combined strength sums the two categories’ average positive correlations. Ranges are simulation percentiles, not confidence intervals for observed means. Every displayed observed statistic exceeds all corresponding simulations.

Shares based on residual signs, however, are not unusual relative to their simulations: all 12 two-sided probabilities are at least 0.164. A share near one-half can therefore coexist with strength beyond the specified benchmark. This is evidence about aggregate statistical relationships, not the economic share of genuine transmission or the significance of individual links. Some differences between the two categories’ strengths depart from the benchmark even with residual signs, so exact symmetry is not established (Appendix C).

4 Discussion and conclusion

The central finding is that factor adjustment and shock classification make distinct contributions to measured downside dominance. Keeping original signs reveals substantial attenuation with a remaining negative imbalance. Classifying shocks relative to the factor benchmark brings the share

close to one-half, while residual network strength remains above the specified simulation benchmark. Empirical applications should report the sign definition and the intermediate comparison with original signs retained.

Several limits qualify this interpretation. Common factors can themselves contain disturbances originating in industries; filtering can remove part of a propagated shock. Conversely, unmeasured common shocks or volatility changes can contribute to remaining dependence. Statistical changes therefore do not identify contagion (Forbes and Rigobon, 2002).

The resampling intervals treat estimated networks as given, and their main calculation uses only two long blocks of consecutive estimates. Simulations repeat estimation but keep the observed factor history and fitted industry volatility parameters fixed. The available simulation summary does not include the complete run settings or industry-level model-fit diagnostics, limiting assessment of the benchmark. Finally, the checks followed inspection of earlier results. The evidence is exploratory and supports a measurement finding, rather than a causal account or proof of exact symmetry.

A Measurement and conditional intervals

The common daily panel has 13,617 observations from December 29, 1971 through December 31, 2025. Removing incomplete boundary weeks yields 2,817 weekly observations from January 7, 1972 through December 26, 2025. Holiday weeks retain observed sessions without rescaling. Each factor model uses its corresponding published factor construction. For industry i , day d , and factor model m , rolling regressions give

$$u_{i,d}^m = r_{i,d} - r_{f,d} - \hat{\alpha}_{i,d}^m - (\hat{\beta}_{i,d}^m)' f_d^m. \quad (\text{A.1})$$

All archived coefficient fits end before their application blocks. For week w , define

$$V_{i,w}^m = \sum_{d \in w} (u_{i,d}^m)^2, \quad (\text{A.2})$$

$$V_{i,w}^{m,-,\text{fixed}} = \sum_{d \in w} (u_{i,d}^m)^2 \mathbf{1}\{r_{i,d} < 0\}, \quad (\text{A.3})$$

$$V_{i,w}^{m,-,\text{own}} = \sum_{d \in w} (u_{i,d}^m)^2 \mathbf{1}\{u_{i,d}^m < 0\}, \quad (\text{A.4})$$

$$V_{i,w}^{m,+g} = V_{i,w}^m - V_{i,w}^{m,-g}. \quad (\text{A.5})$$

Here fixed means original signs, own means residual signs, and g indexes the sign rule. Applying analogous definitions to raw and excess returns gives nine configurations: raw plus two sign rules for excess returns and each of the three factor models.

The baseline uses $L_\delta(v) = \log \max(v, \delta)$, with $\delta = 10^{-12}$ in squared-decimal-return units. The receiver-HAR response is $Y_{i,w}^m = L_\delta(V_{i,w+1}^m)$; the source predictor is $X_{j,w}^{m,s,g} = L_\delta(V_{j,w}^{m,s,g})$. Controls are an intercept, current log total variation, logs of four- and thirteen-week arithmetic averages of total variation ending at w , and the archived market log-variation series lagged one source week. Averages precede logarithms. The same market control enters every return configuration.

At update t , source weeks $t - 156, \dots, t - 1$ are paired with responses through t . Let $M_{i,t}^m$ remove the receiver's controls by least squares. For $i \neq j$,

$$a_{ij,t}^{m,s,g} = \frac{(M_{i,t}^m Y_i^m)' (M_{i,t}^m X_j^{m,s,g})}{\|M_{i,t}^m Y_i^m\| \|M_{i,t}^m X_j^{m,s,g}\|}. \quad (\text{A.6})$$

For $N = 30$, suppressing configuration superscripts,

$$S_t^s = \frac{\sum_{i \neq j} \max(a_{ij,t}^s, 0)}{N(N-1)}, \quad T_t = S_t^- + S_t^+, \quad A_t = S_t^- - S_t^+, \quad q_t = \frac{S_t^-}{T_t}. \quad (\text{A.7})$$

The reported share is $\bar{q} = n^{-1} \sum_t q_t$, not $\bar{S}^- / (\bar{S}^- + \bar{S}^+)$. The sequential comparison satisfies

$$\bar{q}^{\text{raw}} - \bar{q}^{m,\text{own}} = (\bar{q}^{\text{raw}} - \bar{q}^{m,\text{fixed}}) + (\bar{q}^{m,\text{fixed}} - \bar{q}^{m,\text{own}}). \quad (\text{A.8})$$

This is an accounting identity for the specified sequence of choices.

Conditional intervals use 2,000 paired circular moving-block draws with seed 20260912, following the dependent-data rationale of Künsch (1989). Blocks of 26, 52, and 78 unique updates were considered; reported intervals use 78. The same resampled dates enter each contrast. Neither factors nor networks are refitted in this bootstrap. Adjacent updates share 152 response weeks, and only two 78-update blocks fit within the evaluation sample. Population coverage is not established; the separate simulation experiment does not validate that coverage.

B Transformation checks

Table B.1 reports all eight observed treatments. Alternative floors use the same logarithmic rule with $\delta = 10^{-10}, 10^{-8}, 10^{-6}$. Smooth alternatives use

$$L_{\lambda,t}^m(v) = \log(v + \lambda M_t^m), \quad \lambda \in \{0.01, 0.10\}, \quad (\text{B.1})$$

where M_t^m is median total weekly variation pooled across industries and the 156 source weeks in the estimation window. The same model-specific scale applies to signed sources, responses, and own-volatility controls under both sign rules. The market control uses its own window median. Future response observations do not determine the scales.

Ranks replace each source, response, and nonconstant control by its within-window rank, averaging ties, before residualization. Zero-week exclusion retains a source week only if both source categories have positive variation in all nine configurations. The source-specific mask is shared by both categories and all receivers; controls are refitted on retained rows without compressing the calendar or recomputing historical averages on the reduced sample. Sources with fewer than 32 retained rows are unavailable, and both categories use the same finite off-diagonal edge set.

Table B.1: Negative shares across transformations (%)

Treatment	Raw	Market		FF3		FF5	
		Original	Residual	Original	Residual	Original	Residual
Baseline 10^{-12}	94.37	63.64	48.95	62.76	49.16	62.84	49.14
Floor 10^{-10}	95.15	63.45	48.89	62.86	49.27	62.96	49.41
Floor 10^{-8}	95.90	62.89	48.68	62.89	49.45	63.07	49.78
Floor 10^{-6}	96.34	61.66	48.98	62.23	49.99	62.81	50.41
Offset $0.01M$	96.28	61.32	48.94	62.07	50.01	62.65	50.48
Offset $0.10M$	95.42	59.52	49.21	60.57	50.32	61.32	51.11
Ranks	95.08	59.15	48.96	60.29	50.60	60.86	51.35
Zero exclusion	93.01	58.25	48.57	58.74	49.47	59.44	49.93

Original and residual refer to the daily sign definitions. Entries average 156 update-specific ratios. Excess-return controls are included in matching but omitted from the table. All transformed variables are changed together. Zero exclusion changes the within-window sample. These are point estimates, not equivalence tests.

C Simulation design and reporting scope

For each generating factor model, retain observed factors, the risk-free rate, the archived market control, and fitted common returns $c_{i,d}^m = \widehat{\alpha}_{i,d}^m + (\widehat{\beta}_{i,d}^m)' f_d^m$. Industry residuals follow a zero-mean GJR-GARCH(1,1) process (Glosten et al., 1993):

$$e_{i,d}^* = \sqrt{h_{i,d}^*} z_{i,d}^*, \quad (\text{C.1})$$

$$h_{i,d}^* = \omega_i + \lambda_i (e_{i,d-1}^*)^2 + \gamma_i (e_{i,d-1}^*)^2 \mathbf{1}\{e_{i,d-1}^* < 0\} + \phi_i h_{i,d-1}^*. \quad (\text{C.2})$$

Parameters are estimated separately by Gaussian quasi-maximum likelihood. Innovations are drawn from each industry's centered, unit-variance empirical standardized-residual pool, independently

across industries and dates. The simulation therefore allows non-Gaussian shocks and asymmetric volatility without direct residual links. The asymmetry coefficient is not restricted to be positive, subject to variance positivity and stationarity restrictions.

Simulated raw returns are $r_{i,d}^* = r_{f,d} + c_{i,d}^m + e_{i,d}^*$. Every draw refits the 504-day factor regressions at the original 21-day update dates and rebuilds weekly variation and networks. Original signs within a simulation mean its own simulated raw-return signs, not the observed signs. Each generating model has 499 draws, covering the baseline, both smooth offsets, and ranks. Alternative hard floors and zero exclusion are not simulated.

For a statistic Q , the upper-tail reference probability is

$$p_{\text{upper}} = \frac{1 + \sum_{b=1}^{499} \mathbf{1}\{Q_b^* \geq Q_{\text{obs}}\}}{500}. \quad (\text{C.3})$$

The lower tail replaces $\geq$ by $\leq$; the two-sided value is $\min\{1, 2 \min(p_{\text{lower}}, p_{\text{upper}})\}$. These are unadjusted, model-based Monte Carlo summaries. A value of 0.002 means no simulated statistic reached the observed value, not that the underlying tail probability is known exactly. The reference distributions condition on fitted parameters and common paths.

Table C.1: Raw negative shares under the simulation benchmark (%)

Model	Observed	Simulation mean	95% range	Two-sided p
Market	94.37	96.39	[95.09, 97.58]	0.012
FF3	94.37	94.96	[93.28, 96.38]	0.408
FF5	94.37	94.70	[93.34, 96.07]	0.660

Baseline transformation; 499 draws per model. The market-model lower-tail probability is 0.006. Ranges are simulation percentiles.

The raw-share comparison does not establish overall model fit. Observed raw combined strength, 0.11824, exceeds the FF3 and FF5 reference distributions ($p_{\text{upper}} = 0.002$ for each). Among residual-sign results, the FF5 strength difference $\bar{A}$ is positive under the 10% offset and ranks, with two-sided probabilities 0.016 and 0.028. These exceptions qualify any claim of exact residual symmetry. Separate comparisons of statistic levels also do not test the paired raw-to-original-sign or original-to-residual-sign changes.

The implementation recovers factor paths from the archived full-rank loading matrices and checks common-return reconstruction and historical rolling-regression replication to 10^{-10} in decimal returns. Archived coefficients initialize days lacking sufficient prior history; network windows and their controls must follow fully re-estimated residuals. Marginal estimation uses three feasible starts and retains the best converged fit. It requires $\omega_i > 0$, $\lambda_i \geq 0$, $\lambda_i + \gamma_i \geq 0$, and $\phi_i \geq 0$, and separately checks stationarity under the empirical innovation pool. Invalid fits or paths stop computation. Numerical positive scores at or below 10^{-12} and degenerate residual correlations are assigned zero using the archived tolerance rules.

The implementation defaults to a Gaussian-persistence cap of 0.995, 2,000 burn-in days, seed 20260915, and calibration through the last requested network update. These configurable defaults cannot be verified as the full-run settings from the available summary, which confirms the simulation count and transformations but omits the run manifest and industry-level fit diagnostics. Adequacy of simulated variances, tails, sign frequencies, and volatility persistence therefore remains to be

documented. Rejection of this benchmark does not rule out other models without direct industry transmission.

Declaration of generative AI use

ChatGPT (OpenAI) assisted with condensing and restructuring the manuscript and preparing the LaTeX source. The author is responsible for final verification and all manuscript content.

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